**11.** An ideal gas is confined in a cylinder fitted with a frictionless piston. When 90 J of heat is supplied and the gas is allowed to expand $2.5 \times 10^{-4}$ m$^3$ in volume at a constant pressure of $2.0 \times 10^5$ Pa, the temperature rises by 4.5 K.

What is the increase in internal energy during this process?

**A** 40 J **B** 50 J **C** 90 J **D** 140 J

**Ans: A**

Expanding under constant pressure:

$$\begin{aligned}
\Delta U &= Q + W_{on} \\
&= Q + (-p\Delta V) \\
&= 90 + (-2.0 \times 10^5 \times 2.5 \times 10^{-4}) \\
&= 90 - 50 \\
&= 40 \text{ J}
\end{aligned}$$

**12** The graph below shows the variation with time of the velocity of a 5.0 kg oscillating